EX:4 WRITE THE PROGRAM FOR CRYPT-ARITHMETIC PROBLEM

Aim

To develop a Python program that solves a **Crypt-Arithmetic Problem** by assigning unique digits to letters such that the given arithmetic equation is satisfied.

ALGORITHM

Step 1: Start the program.

Step 2: Read the first word, second word, and result word from the user.

Step 3: Extract all unique letters from the three words.

Step 4: Check whether the number of unique letters is greater than 10. If yes, display **"Too many unique letters"** and stop.

Step 5: Generate all possible permutations of digits (0–9) for the unique letters.

Step 6: Assign each unique letter a unique digit from the generated permutation.

Step 7: Ensure that the first letter of each word is not assigned the digit **0**.

Step 8: Convert the three words into their corresponding numerical values using the assigned digits.

Step 9: Check whether the equation **Word1 + Word2 = Result** is satisfied.

Step 10: If the equation is true, display the solution and the letter-to-digit mapping, then stop.

Step 11: If no valid assignment is found after checking all permutations, display **"No Solution Found"**.

Step 12: End the program.

PROGRAM CODE:

```
from itertools import permutations


word1 = input("Enter First Word: ").upper()

word2 = input("Enter Second Word: ").upper()

result = input("Enter Result Word: ").upper()
```

```
letters = ""

for ch in word1 + word2 + result:
    if ch not in letters:
        letters += ch

if len(letters) > 10:
    print("Too many unique letters.")
    exit()

digits = "0123456789"

for p in permutations(digits, len(letters)):
    d = dict(zip(letters, p))

    if d[word1[0]] == '0' or d[word2[0]] == '0' or d[result[0]] == '0':
        continue

    n1 = int("".join(d[ch] for ch in word1))
    n2 = int("".join(d[ch] for ch in word2))
    n3 = int("".join(d[ch] for ch in result))

    if n1 + n2 == n3:
        print("\nSolution Found")
        print(word1, "=", n1)
        print(word2, "=", n2)
        print(result, "=", n3)
```

```
        print("\nLetter Mapping")
        for k, v in d.items():
            print(k, "=", v)
        break
else:
    print("No Solution Found")
```

OUTPUT:

```
>>
====== RESTART: C:/Users/Hp/AppData/Local/Programs/Python/Python314/EX4.py ==
Enter First Word: send
Enter Second Word: more
Enter Result Word: money

Solution Found
SEND = 9567
MORE = 1085
MONEY = 10652

Letter Mapping
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
>>
```

RESULT

The Python program successfully solves the Crypt-Arithmetic problem by finding a valid digit assignment for each letter.The solution and the corresponding letter-to-digit mapping are displayed if a valid assignment exists.